

Walmart Sales Data Analysis and Visualization Using Power BI

Report by:

JENIFAR JOSEMINA S
B.COM FINTECH AND AI
E25FA014

INTRODUCTION:

Walmart is one of the world's largest retail companies, serving millions of customers through its stores and online platforms. With a wide range of products and a large volume of transactions, Walmart generates significant amounts of sales and customer-related data. Analysing this data helps understand sales performance, customer demand, product categories, regional performance, and changes in sales over time. Therefore, data analysis plays an important role in supporting effective business decisions.

Microsoft Power BI is a business intelligence and data visualization tool that helps transform raw data into meaningful and interactive reports. In this project, the Walmart sales dataset is analysed using Power BI to identify important patterns and trends in the data. The data is cleaned and transformed using Power Query, while DAX can be used to create calculated measures for analysing key performance indicators such as total sales, profit, and sales growth. Various charts, graphs, cards, and filters are used to present the analysis in an easy-to-understand manner.

The main purpose of this project is to develop an interactive Power BI dashboard that provides a clear view of Walmart's sales performance. The analysis helps identify high-performing categories, regions, products, and time periods, while also highlighting areas that may require attention. By converting raw Walmart data into visual insights, the project demonstrates how Power BI can support data-driven decision-

making and improve the understanding of business performance.

OBJECTIVES:

- To analyse Walmart sales data using Microsoft Power BI.
- To clean, transform, and organize the Walmart dataset for analysis.
- To identify sales trends and patterns across different time periods.
- To analyse sales performance based on products, categories, and regions.
- To calculate and monitor key performance indicators such as Total Sales, Profit, and Sales Growth.
- To create an interactive dashboard using charts, graphs, cards, and filters.
- To generate meaningful business insights that can support data-driven decision-making.

DATASET DESCRIPTION:

The dataset contains **weekly sales records from Walmart stores**, along with economic and seasonal factors that may influence sales performance. Each record represents the sales information for a particular store during a specific week.

Key Fields in the Dataset

- **Store** – Unique identification number assigned to each Walmart store, ranging from **1 to 45**.
- **Date** – Represents the specific week for which the sales data was recorded.
- **Weekly Sales** – Total sales generated by a particular store during the given week. This is the primary measure used to evaluate sales performance.
- **Fuel Price** – Average fuel price during the respective week, which can be analysed to understand its possible relationship with sales.
- **CPI (Consumer Price Index)** – Indicates the level of changes in consumer prices and helps analyse the impact of economic conditions on sales.
- **Unemployment** – Represents the unemployment rate during the particular week and provides an economic indicator for analysing changes in consumer spending.
- **Holiday Flag** – Indicates whether the particular week included a holiday, helping to analyse the effect of holiday periods on Walmart's weekly sales.

DATA PREPARATION

- **Imported the Walmart dataset into Power BI Desktop** for further analysis and visualization.

- **Corrected the Date data type** so that Power BI could properly recognize the weekly records and support time-based analysis.
- **Verified numerical fields** such as Weekly Sales, Fuel Price, CPI, and Unemployment to ensure that they were correctly recognized for calculations.
- **Checked the Store and Holiday Flag fields** to ensure that they could be used effectively for filtering and comparison.
- **Reviewed the dataset for data consistency** before creating visuals and calculations.
- **Prepared the data for DAX calculations**, allowing key performance indicators to be displayed in the dashboard.

DAX MEASURES USED

Four key DAX measures were created to support the **KPI cards** in the dashboard:

- **Total Sales** – Calculates the overall sales generated across all stores and weeks, providing a high-level view of Walmart's total sales performance.
- **Average Sales** – Calculates the average weekly sales value, helping to understand the typical sales level across the dataset.

- **Store Count** – Identifies the total number of unique Walmart stores included in the dataset, allowing the analysis to cover the complete store network.



- **Maximum Sales** – Identifies the highest weekly sales value recorded, helping to highlight the period with the strongest sales performance.

DASHBOARD COMPONENTS

The dashboard combines KPI cards, charts, and filters to provide a comprehensive view of weekly sales performance.

- ✓ **Total Sales – 6.74 Billion:**

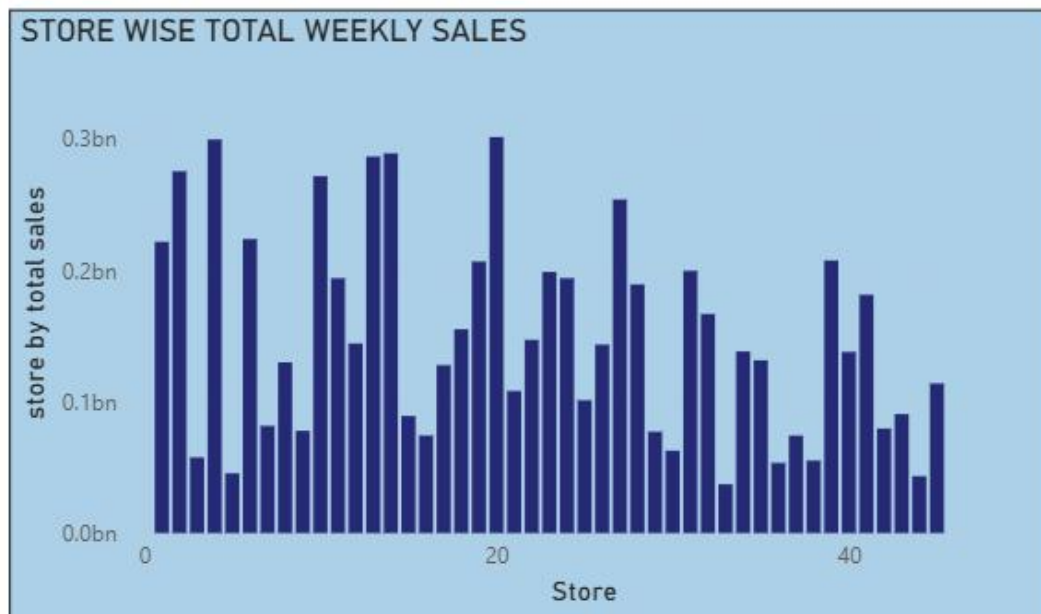
Displays the total sales generated across the dataset. It provides an overall view of Walmart's sales performance during the analysed period.

- ✓ **Average Sales – 1.05 Million:**

Represents the average sales value across the available records. This helps understand the typical sales level within the dataset.

✓ **Store Count – 6K:**

Displays the count of store-related records considered in the analysis. It gives an overview of the number of store records represented in the dataset.



✓ **Transaction Count – 6K:**

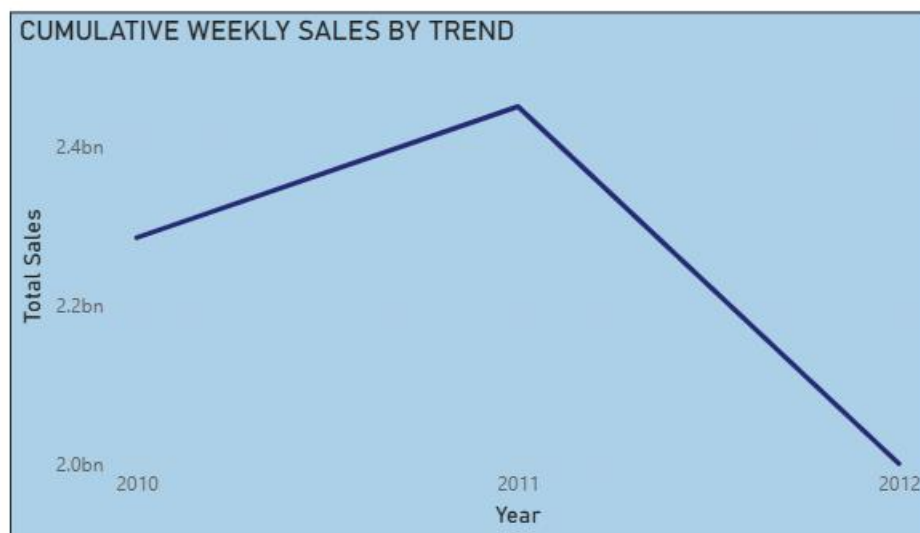
Shows the total number of transaction records included in the analysis. It helps indicate the volume of records used to generate the dashboard insights.

STORE WISE TOTAL WEEKLY SALES:

A column chart is used to compare the total weekly sales across different Walmart stores. The Store is represented on the X-axis, while total weekly sales are represented on the Y-axis.

- The chart allows a store-by-store comparison of sales performance.
- It highlights differences in sales levels among the 45 stores.

- Stores with higher bars represent higher total weekly sales during the analysed period.
- Stores with lower bars indicate comparatively lower total weekly sales.



- The visualization makes it easier to identify variations in store performance.
- It provides a clear overview of how sales are distributed across Walmart's store network.

WEEKLY SALES TREND:

A line chart is used to show the cumulative weekly sales trend across different years. The year is represented on the X-axis, while total sales are represented on the Y-axis.

- The chart shows the cumulative sales trend for 2010, 2011, and 2012.

- Sales increased from approximately 2.3 billion in 2010 to around 2.45 billion in 2011.
- The chart shows a decline to approximately 2.0 billion in 2012.



- The upward movement between 2010 and 2011 indicates an increase in cumulative sales.
- The downward movement from 2011 to 2012 shows a reduction in the cumulative sales value represented in the dataset.
- The line chart makes it easy to observe year-wise changes and overall sales movement

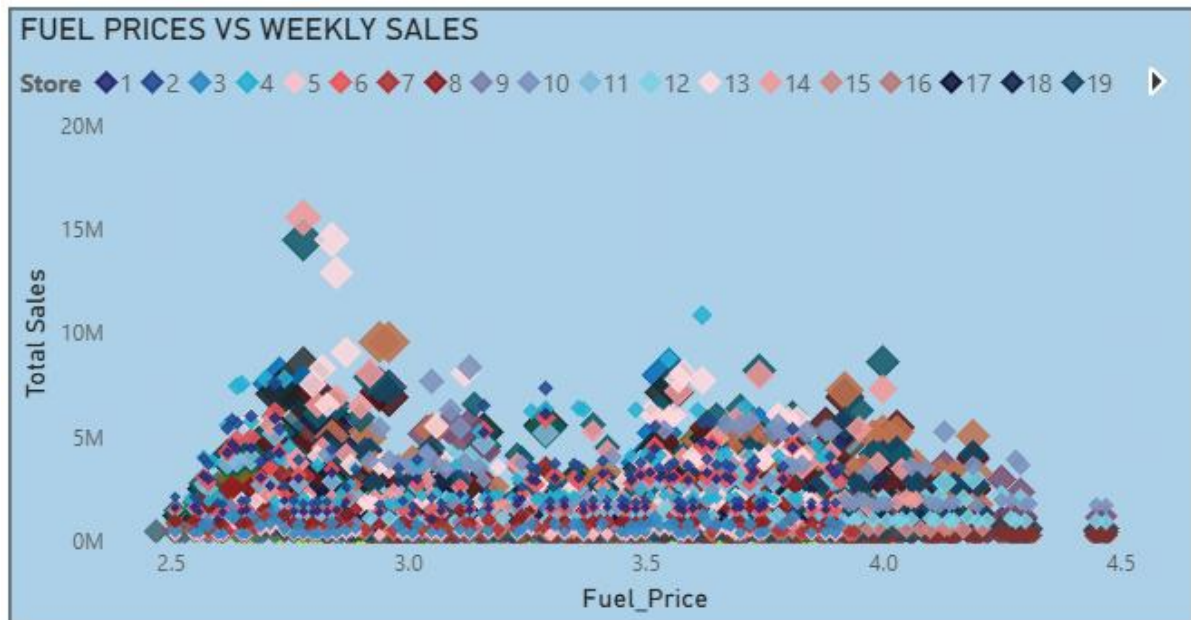
AVERAGE WEEKLY SALES BY STORE:

A horizontal bar chart is used to compare the average weekly sales across different Walmart stores. The Store is represented on the Y-axis, while Average Sales is represented on the X-axis.

- The chart shows the average weekly sales for each store during the analysed period.
- It allows direct comparison of the typical sales performance of individual stores.
- Stores with longer bars have higher average weekly sales.
- Stores with shorter bars have comparatively lower average weekly sales.
- The chart shows noticeable variation in average sales among the stores.
- Using average sales helps provide a balanced view of store performance rather than relying only on total sales.

Fuel Price vs Weekly Sales

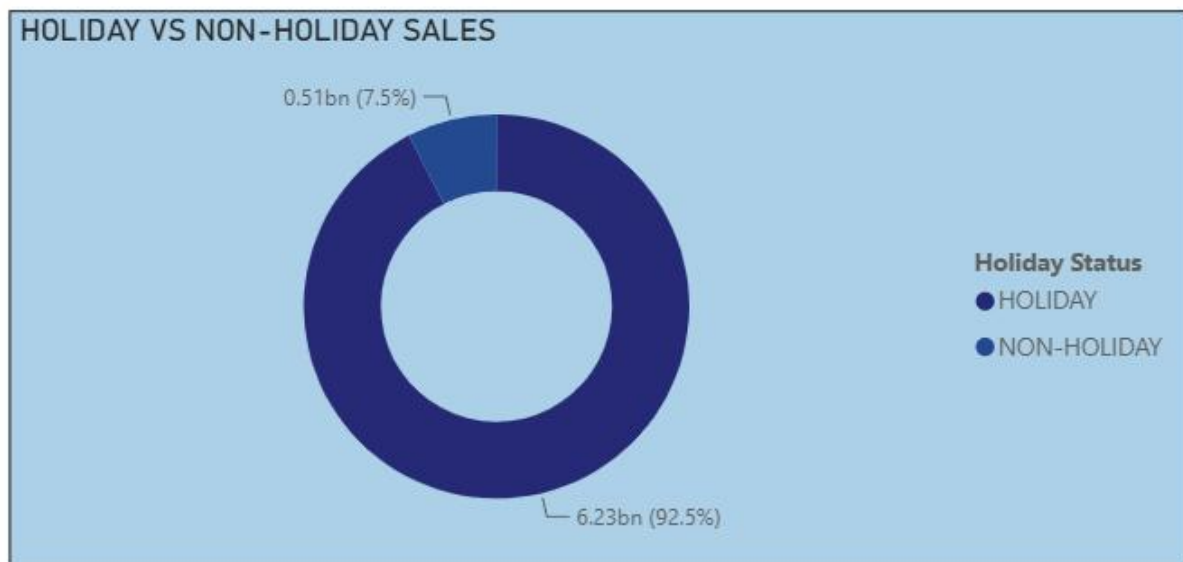
A scatter chart is used to analyse the relationship between fuel prices and weekly sales across different Walmart stores.



- The chart compares fuel price levels with weekly sales across the stores.
- It shows that sales values vary considerably at different fuel price levels.
- Most observations are concentrated at relatively lower sales values, while a few observations show higher sales peaks.
- The distribution of points does not show a simple, consistent upward or downward pattern between fuel price and sales.
- The Store legend allows individual stores to be distinguished and compared within the same visual.

- The chart helps explore whether changes in fuel prices are associated with changes in Walmart's weekly sales.

HOLIDAY VS NON-HOLIDAY SALES:



A **doughnut chart** is used to compare Walmart's total sales during **holiday and non-holiday periods**. The chart presents the contribution of each holiday status to the overall sales value.

- **Non-Holiday Sales:** Approximately **6.23 billion (92.5%)**, representing the majority of the total sales.
- **Holiday Sales:** Approximately **0.51 billion (7.5%)**, representing a smaller portion of the total sales.
- The chart clearly shows the difference between sales recorded during holiday and non-holiday periods.

- Non-holiday weeks account for a significantly larger share of the total sales in this dataset.
- The visual provides a simple way to understand the distribution of sales according to holiday status.

The image displays three stacked filter slicers from a Power BI dashboard, all with a light purple background and dark text.

- FILTERS STORE:** The top slicer has a title bar labeled "FILTERS" and "STORE". Below it is a dropdown menu showing "All" with a downward arrow icon.
- DATE:** The middle slicer has a title bar labeled "DATE". It features a date range selector with two input boxes: "05-02-2010" and "26-10-2012", each with a calendar icon. Below these is a horizontal timeline with two white circular handles.
- HOLIDAY STATUS:** The bottom slicer has a title bar labeled "HOLIDAY STATUS". It contains two checkboxes: "HOLIDAY" and "NON-HOLIDAY", both of which are currently unchecked.

SLICERS USED:

Slicers are interactive filtering components used in the Power BI dashboard. They allow users to select specific

conditions and dynamically update the dashboard visuals according to the selected criteria.

- **Store Slicer:**

Allows users to select a specific Walmart store or view the data for all stores. This helps analyse and compare the sales performance of individual stores.

- **Date Slicer:**

Allows users to select a specific date or a particular time period using the date range slider. This helps examine sales performance over a selected period.

- **Holiday Status Slicer:**

Provides options to filter the data based on **Holiday** or **Non-Holiday** periods. This helps compare sales performance under different holiday conditions.

CONCLUSION:

The Walmart Sales Analysis project demonstrates how Power BI can transform raw sales data into meaningful and interactive business insights. Through KPI cards, charts, and slicers, the dashboard provides a clear view of sales performance across stores, time periods, fuel prices, and holiday status. Overall, the project highlights the usefulness of data visualization and business intelligence in understanding sales patterns and supporting data-driven decision-making.